Jason Chen

(267)-721-5370 | jwtchen@seas.upenn.edu | jason2504.github.io | linkedin.com/in/jason-chen

Education

University of Pennsylvania (GRASP Lab) Aug. 2024 – May. 2026
MS in Mechanical Engineering – Mechatronics and Robotic Systems, GPA: 4.0/4.0 *Philadelphia, PA*

- **Relevant Coursework**: Design of Mechatronic Systems, Introduction to Robotics, Finite Element Analysis, Machine Learning, Distributed Robotics, Materials and Manufacturing for Mechanical Design, Integrated CAD Design

University of Alberta Sep. 2019 – Apr. 2024
BS in Mechanical Engineering Co-op – Plan IV, GPA: 3.9/4.0 *Edmonton, AB*

- **Certificates**: Certified SOLIDWORKS Professional (CSWP), Fundamentals of Engineering Mechanical

Experience

Figueroa Robotics Lab | Graduate Research Assistant Apr. 2025 – Present
Dual-Mode Robotic Systems, CAD Design, Mechanical Prototyping *Philadelphia, PA*

- Designed and prototyped two dual-mode robotic systems with 4-DOF per arm capable of both precision manipulation and dynamic locomotion: rigid joint-based system with custom universal joints and servo mechanisms, and cable-actuated continuum arm with distributed motor control and flexible vertebrae
- Engineered innovative actuation systems including timing belt transmission with bevel gear coupling for universal joints, and cooperative cable control where adjacent motors jointly actuate spine segments for enhanced redundancy
- Developed compliant 3-finger gripper with worm gear reduction and spring-based adaptive control, enabling seamless transitions between delicate manipulation and robust ground contact support

ATCO Ltd., Natural Gas Division | Project Services Engineering Intern Sep. 2022 – Sep. 2023
Process Automation, Power Platform, Project Management *Edmonton, AB*

- Engineered automated workflow solutions using Microsoft Power Platform (Power Apps, Power Automate & Power BI), transforming manual processes into digital applications with 85% reduction in processing time
- Supervised and mentored 4 engineering interns, implementing standardized documentation protocols for 20+ projects
- Performed cost analysis with Primavera P6, optimizing supply chain and scheduling for $15M+ capital projects
- Designed metadata-driven SharePoint system, reducing document retrieval time by 70% for 20+ project teams

Advanced Composite Material Engineering Group | R&D Assistant Jan. 2022 – Sep. 2022
Thermal Analysis, FEA, Prototyping, WHO Testing Protocols *Edmonton, AB*

- Engineered a prototype long-range vaccine cold box achieving -78.5°C to 8°C temperature maintenance for COVID-19 vaccine transportation in Mozambique (prototype was field tested and successfully deployed in Mozambique in 2023)
- Performed finite element analysis (FEA) and thermal simulations in SolidWorks to optimize material selection and structural integrity under 1-meter drop test conditions
- Fabricated prototype using polyurethane casting with glass bead reinforcement and developed specifications for injection molding scale-up
- Executed WHO-compliant thermal testing protocols in environmental chamber, achieving 168-hour cold life

Projects

Autonomous Mobile Robot Platform | *Project Manager* Jan. 2024 – Apr. 2024

- Led a team of 5 graduating engineers in designing an AMR platform supporting a 6-DOF Yaskawa GP-50 robotic arm for industrial metal additive manufacturing
- Designed tracked mobility system capable of 2-ton payload, 15% grade traversal, and extreme temperature operation (-40°C to 30°C) with 6-hour runtime
- Conducted FEA analysis using SolidWorks to validate structural integrity with 160-microns end-tool deflection and maximum stress of 62 MPa (safety factor of 5.7)

Mario Kart Transmission Design Project | *Mechanical Team Member* May 2021 – Aug. 2021

- Designed a 5-speed + reverse sequential dog-clutch transmission system for high-speed racing applications
- Calculated bearing load ratings, spline/keyway stress distributions, and torque transmission paths using analytical and computational methods
- Specified material properties and tolerance requirements (±0.2mm) for precision components to ensure durability

Technical Skills

Computer Aided Design & Simulation: SolidWorks, Star-CCM+, DaVis, COMSOL, URDF Modeling, Keyshot
Programming & Robotics: Python, MATLAB, Simulink, C, ROS, Google Colab, Git, Linux, ESP32/Arduino, AprilTag/Computer Vision, PWM Motor Control, CAN Bus Communication
Manufacturing & Prototyping: 3D Printing (FDM), Laser Cutting, Injection Molding, Polyurethane Casting, CNC Mill, Composite Fabrication, Mastercam, GD&T
Automation & Data Tools: SharePoint, Power BI, Power Apps, Power Automate, MS Office